The pH scale

Litmus paper can only tell you if a substance is an **acid** or a **base**. It cannot tell you how strong an acid or base it is. This is what tells you how dangerous it is. **Universal indicator** is a special indicator that can be used to measure the strength of an acid or base on the **pH scale**. The indicator changes colour depending on the strength of the acid or base.

☞ **1** Use the words in the word bank to complete the passage.

You can describe how a_____ or how b_____ a substance is by using the numbers on the pH s_____. The pH scale ranges from 1 to 14. Numbers l_____ than pH 7 mean that substances are acidic. Numbers h_____ than pH 7 mean that the substance is basic. If a substance has a pH of 7, it is n_____ – neither acidic nor basic. Substances with a pH of 1 are s_____ acids and substances with a pH of _____ are strong bases.

Word bank
14
scale
acidic
neutral
basic
higher
lower
strong

2 Colour in and label the pH scale below.

Deep red	Red	Pink	Orange	Yellow-orange	Yellow	Green	Dark green	Blue-green	Blue	Light purple	Purple	Dark purple	Darker purple

pH 1 2 3 4 5 6 7 8 9 10 11 12 13 14

s_____ w_____ n_____ w_____ s_____

a_____ a_____ b_____ b_____

3 Circle the correct ending for each of the sentences below.

A substance that is pH 13 is	a strong acid.	a weak base.	a strong base.
A substance with a pH of 7 will turn	purple.	green.	red.
Acids have a pH	less than 7.	between 5 and 9.	between 8 and 14.

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